

National Academy of Science and Technology
(Affiliated to Pokhara University)

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Pre-University Examination

Semester : VII_Fall

Level: Bachelor
Program: B.E. Computer
Course: Network and Cyber Security

Year : 2025
F.M. : 100
P. M. : 45
Time : 3 Hrs.

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

- 1) a) What are the different types of attack performed in network? 7
Explain the network security access model.
- b) Define cryptography and cryptology. Explain the general depiction of DES. 8
- 2) a) Explain concept of public key cryptography. Explain mixed column of AES with example. 7

Or

Explain the encryption decryption process of playfair cipher.

Encrypt the message Instrument using key monarchy. *GAATSMZLWRA*

- b) Define substitution and transposition cipher. Encrypt the message "HOPE" using hill cipher with key $\begin{matrix} 7 & 8 \\ 11 & 11 \end{matrix}$ 8
TSVW

- 3) a) What is concept of prime factorization problem? Using RSA, choose $p = 11$ and $q = 7$, encode the message 7. 7
- b) Explain Diffie-Hellman key exchange algorithm. Verify if 2 is primitive root of 11 or not 8

Or

Alice publishes the ElGamal public key $(q, \alpha, Y_A) = (23, 5, 8)$. Bob desired to send the secret message LOCK to Alice. Using the equivalence $A = 0, B=1, \dots, Z=25$, encrypt the message using Alice's public key. Use a random number $k = 4$. Also find the private key of Alice.

- 4) a) How does the n^2 key distribution problem can be solved? 7
b) What do you mean by digital signature? How it can be achieved? 8
- 5) a) Define hash code. Explain the components of digital certificate. 7
b) What do you mean by SSL? Explain SSL protocol stacks. 8
- 6) a) How does PGP ensure the email security? 7
b) Why data protection is needed? How does the server side request forgery can be prevented? 8
- 7) Write short notes on any two: 2×5
a) HMAC
b) IPSec Transport Mode
c) Types of injection